

Capitolul 2
Problema teoretica

2.20 Fie A o matrice patratica diagonal dominanta. Aratati ca:

- a) A este nesingulara
b) Metoda lui Jacobi pentru sistemul $Ax=b$ converge pentru orice b

Punctul a)

Deoarece A este o matrice patratica diagonal dominanta $\Rightarrow \forall i \in \{1, 2, \dots, n\} |a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$.

Vom folosi metoda reducerii la absurd.

Presupunem ca matricea A este singulara $\Rightarrow \exists x \neq 0$ ai $Ax = 0$

$$\text{Fie } x = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} \quad \forall i \in \{1, 2, \dots, n\} \quad \sum_{j=1}^n a_{ij}x_j = 0 \Rightarrow a_{ii}x_i = - \sum_{j=1, j \neq i}^n a_{ij}x_j$$

Se stie ca $\|x\|_{\infty} = \max\{|x_1|, \dots, |x_n|\}$.

Alegem x ai $\|x\|_{\infty} = 1$ si fie $\|x\|_{\infty} = |x_i|$.

Notam cu $x_{\max} = \max\{|x_1|, \dots, |x_n|\} = \|x\|_{\infty} = |x_i| = 1$.

$$\text{Deoarece } a_{ii}x_i = - \sum_{j=1, j \neq i}^n a_{ij}x_j \Rightarrow |a_{ii}x_i| = \left| \sum_{j=1, j \neq i}^n a_{ij}x_j \right| \Rightarrow |a_{ii}x_i| \leq \sum_{j=1, j \neq i}^n |a_{ij}x_j|$$

$$\Rightarrow |a_{ii}| |x_i| \leq \sum_{j=1, j \neq i}^n |a_{ij}| |x_j| \quad | : x_{\max}$$

$$\Rightarrow |a_{ii}| |x_i| / x_{\max} \leq \sum_{j=1, j \neq i}^n |a_{ij}| |x_j| / x_{\max} \Rightarrow |a_{ii}| \leq \sum_{j=1, j \neq i}^n |a_{ij}| \quad (\text{rel 1})$$

$\downarrow$
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$$\text{Deoarece } \|x\|_{\infty} = 1 \Rightarrow |x_j| \leq 1 \Rightarrow \sum_{j=1, j \neq i}^n |a_{ij}| |x_j| \leq \sum_{j=1, j \neq i}^n |a_{ij}| \quad (\text{rel 2})$$

Din (rel 1) si (rel 2) $\Rightarrow |a_{ii}| \leq \sum_{j=1, j \neq i}^n |a_{ij}| \Rightarrow$ contradictie cu definitia matricei patraticice diagonal dominante $\Rightarrow$ matricea A este nesingulara.

Punctul b)

$$Ax = b \quad \text{fie } b = (b_1, b_2, \dots, b_n)$$

$$\forall i \in \{1, \dots, n\} \quad \sum_{j=1}^n a_{ij}x_j = b_i \Rightarrow a_{ii}x_i = b_i - \sum_{j=1, j \neq i}^n a_{ij}x_j \Rightarrow x_i = 1/a_{ii} (b_i - \sum_{j=1, j \neq i}^n a_{ij}x_j)$$

$$\text{Fie } D = \text{diag}(A), L = -\text{tril}(A), U = -\text{triu}(A)$$

$$D = \begin{pmatrix} a_{11} & 0 & 0 & \dots & 0 \\ 0 & a_{22} & \dots & & 0 \\ \dots & & & & \\ 0 & 0 & & & a_{nn} \end{pmatrix} \quad L = -\text{tril}(A) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ -a_{21} & 0 & 0 & 0 & 0 & & \\ -a_{31} & -a_{32} & 0 & 0 & \dots & & \\ \dots & & & & & & \\ -a_{n1} & \dots & \dots & \dots & -a_{nn-1} & 0 \end{pmatrix}$$

$$U = -\text{triu}(A) = \begin{pmatrix} 0 & -a_{12} & -a_{13} & \dots & -a_{1n} \\ 0 & 0 & -a_{23} & \dots & -a_{2n} \\ \dots & & & & \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \quad D^{-1} = \begin{pmatrix} 1/a_{11} & 0 & 0 & \dots & 0 \\ 0 & 1/a_{22} & \dots & & 0 \\ \dots & & & & \\ 0 & 0 & \dots & \dots & 1/a_{nn} \end{pmatrix}$$

$$T = T_J = D^{-1}(L + U) = \begin{pmatrix} 0 & -a_{12}/a_{11} & -a_{13}/a_{11} & \dots & -a_{1n}/a_{11} \\ -a_{21}/a_{22} & 0 & -a_{23}/a_{22} & \dots & -a_{2n}/a_{22} \\ \dots & & & & \\ -a_{n1}/a_{nn} & -a_{n2}/a_{nn} & \dots & \dots & 0 \end{pmatrix}$$

Teorema de convergenta:

Daca $\|T\| < 1$ pentru cel putin o norma matriciala rezulta ca metoda converge.

$$\|A\|_{\infty} = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| \Rightarrow \|T_J\|_{\infty} = \max_{1 \leq i \leq n} \sum_{j=1, j \neq i}^n |a_{ij}/a_{ii}| = \max_{1 \leq i \leq n} \left(\sum_{j=1, j \neq i}^n |a_{ij}| \right) / |a_{ii}| < 1$$

deoarece matricea A este nesingulara adica $\forall i \in \{1, 2, \dots, n\} \quad |a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$.

Din cele de mai sus rezulta ca metoda lui Jacobi converge pentru orice b.